

Sneha Verma

PES1UG23AM309

LAB -6 OOAD

PROBLEM 1:

```
Main.java
1 import java.util.*;
2
3 // Interface
4 interface InsurancePolicy {
5     String getPolicyHolder();
6     double calculatePremium();
7     void displayPolicyDetails();
8 }
9
10 // Health Insurance
11 class HealthInsurance implements InsurancePolicy {
12     String policyHolder;
13     double baseAmount;
14     int age;
15
16     HealthInsurance(String p, double b, int a) {
17         policyHolder = p;
18         baseAmount = b;
19         age = a;
20     }
21
22     public String getPolicyHolder() {
23         return policyHolder;
24     }
25
26     public double calculatePremium() {
27         return baseAmount + (age * 100);
28     }
29
30     public void displayPolicyDetails() {
31         System.out.println("Policy Type: Health Insurance");
32         System.out.println("Policy Holder: " + policyHolder);
33         System.out.printf("Premium: %.2f\n", calculatePremium());
34     }
35 }
36
37 // Vehicle Insurance
38 class VehicleInsurance implements InsurancePolicy {
39     String policyHolder;
40     double vehicleValue;
```

```
39     String policyHolder;
40     double vehicleValue;
41
42     VehicleInsurance(String p, double v) {
43         policyHolder = p;
44         vehicleValue = v;
45     }
46
47     public String getPolicyHolder() {
48         return policyHolder;
49     }
50
51     public double calculatePremium() {
52         return vehicleValue * 0.05;
53     }
54
55     public void displayPolicyDetails() {
56         System.out.println("Policy Type: Vehicle Insurance");
57         System.out.println("Policy Holder: " + policyHolder);
58         System.out.printf("Premium: %.2f\n", calculatePremium());
59     }
60 }
61
62 // Life Insurance
63 class LifeInsurance implements InsurancePolicy {
64     String policyHolder;
65     double coverageAmount;
66     int termYears;
67
68     LifeInsurance(String p, double c, int t) {
69         policyHolder = p;
70         coverageAmount = c;
71         termYears = t;
72     }
73
74     public String getPolicyHolder() {
75         return policyHolder;
76     }
77
78     public double calculatePremium() {
79         return (coverageAmount / termYears) * 0.02;
80     }
81 }
```

```

76 }
77
78 public double calculatePremium() {
79     return (coverageAmount / termYears) * 0.02;
80 }
81
82 public void displayPolicyDetails() {
83     System.out.println("Policy Type: Life Insurance");
84     System.out.println("Policy Holder: " + policyHolder);
85     System.out.printf("Premium: %.2f\n", calculatePremium());
86 }
87 }
88
89 // MAIN CLASS (must be named Main for online compilers)
90 public class Main {
91     public static void main(String[] args) {
92         Scanner sc = new Scanner(System.in);
93
94         int choice = sc.nextInt();
95         sc.nextLine();
96
97         InsurancePolicy policy = null;
98
99         if (choice == 1) {
100             String name = sc.nextLine();
101             double base = sc.nextDouble();
102             int age = sc.nextInt();
103             policy = new HealthInsurance(name, base, age);
104         } else if (choice == 2) {
105             String name = sc.nextLine();
106             double value = sc.nextDouble();
107             policy = new VehicleInsurance(name, value);
108         } else if (choice == 3) {
109             String name = sc.nextLine();
110             double coverage = sc.nextDouble();
111             int term = sc.nextInt();
112             policy = new LifeInsurance(name, coverage, term);
113         }
114
115         policy.displayPolicyDetails();
116     }
117 }
118

```

```

1
sneha
0.9
34
Policy Type: Health Insurance
Policy Holder: sneha
Premium: 3400.90

...Program finished with exit code 0
Press ENTER to exit console.

```

2
SNEHA
0.76
56

Output:

Policy Type: Vehicle Insurance
Policy Holder: SNEHA
Premium: 0.04

PROBLEM 2

GPA.java

+

```
1 import java.util.*;
2
3 class GPA {
4
5     double calculateGPA(int marks) {
6         return marks / 10.0;
7     }
8
9     double calculateGPA(int m1, int m2, int m3) {
10        return (m1 + m2 + m3) / 30.0;
11    }
12
13    double calculateGPA(double percentage) {
14        return percentage / 20.0;
15    }
16
17    double calculateGPA(int marks, int extraCredit) {
18        return (marks + extraCredit) / 10.0;
19    }
20 }
```

```

21
22 public class Main {
23     public static void main(String[] args) {
24         Scanner sc = new Scanner(System.in);
25         GPA g = new GPA();
26
27         int choice = sc.nextInt();
28         double result = 0;
29
30         if (choice == 1) {
31             int marks = sc.nextInt();
32             result = g.calculateGPA(marks);
33         } else if (choice == 2) {
34             int m1 = sc.nextInt();
35             int m2 = sc.nextInt();
36             int m3 = sc.nextInt();
37             result = g.calculateGPA(m1, m2, m3);
38         } else if (choice == 3) {
39             double p = sc.nextDouble();
40             result = g.calculateGPA(p);
41         } else if (choice == 4) {
42             int m = sc.nextInt();
43             int e = sc.nextInt();
44             result = g.calculateGPA(m, e);
45         }
46
47         System.out.printf("%.2f", result);
48     }
49 }
50

```

1
90

STDIN
2
90
92
87

Output:
9.00

Output:
8.97

STDIN
3
180.80

STDIN
4
98
2

Output:
9.04

Output:
10.00